

Statement of Work (SOW)

Continued Support for the NASA SABER Instrument Project

1. Objectives / Requirements

The objective of this task is to provide continued expert support to ensure effective knowledge transfer, technical continuity, and scientific integrity for the Sounding of the Atmosphere using Broadband Emission Radiometry (SABER) instrument, part of the NASA Thermosphere Ionosphere Mesosphere Energetics and Dynamics (TIMED) satellite mission.

The Contractor shall provide:

- Structured mentoring and technical consultation regarding SABER retrieval algorithms and data products.
- Detailed transfer of institutional knowledge related to algorithm development, calibration approaches, data processing methodologies, and dataset evolution.
- Technical guidance in preparation for processing Version 3 of the SABER dataset (covering the full 24-year mission data record).
- Support for evaluation and validation of Version 3 data products.
- Scientific expertise in preparation of documentation and peer-reviewed publications describing updated algorithms and validated data products.

The Contractor shall provide advisory and technical expertise but shall not assume operational control of the TIMED mission or direct responsibility for flight operations.

2. Characteristics, Scope, and Specifications

The work shall consist of scientific, technical, and advisory support activities including:

2.1 Knowledge Transfer

- Mentoring regarding SABER algorithms, retrieval methodologies, and product interrelationships.
- Detailed explanation of uncertainty characterization and appropriate scientific use of SABER data products.
- Historical context regarding calibration decisions and algorithm evolution.

2.2 Version 3 Data Processing Support

- Technical guidance in preparation for the Version 3 reprocessing effort anticipated to begin in late calendar year 2025.
- Advisory support during processing of the full SABER data record (~24 years).
- Consultation on quality assurance and algorithm performance considerations.

Processing of Version 3 is expected to require a minimum of six months and extend into mid-calendar year 2026.

2.3 Validation Support

- Technical consultation regarding validation methodology and comparison with independent datasets.
- Evaluation of uncertainties, biases, and scientific consistency of Version 3 products.

Validation activities are expected to extend through calendar year 2026 and may continue into calendar year 2027.

2.4 Scientific Documentation

- Participation in preparation of journal articles describing validated Version 3 SABER products and updated algorithms.
- Technical review and contribution to documentation supporting data release.

All services provided under this SOW are advisory and scientific in nature and shall be performed in coordination with the SABER Principal Investigator and NASA personnel.

3. Place of Performance

Work under this SOW shall be performed primarily at the Contractor's facilities. Coordination and technical interactions shall occur with NASA Goddard Space Flight Center personnel and the SABER science team through virtual meetings, teleconferences, electronic correspondence, and, if required and separately authorized, occasional in-person meetings.

4. Period of Performance

March 1, 2026 through September 30, 2026.